

Tentamen/Exam

Biometrisk Identifiering / Biometric Recognition 7.50HP

Akademi/School: School of Information Technology (ITE)

Datum/Date: 27 October 2016

Kurskod/Course code: DI2009

Provkod/Test code: I60I

Lärare som skrivit tentan/Teacher: Fernando Alonso-Fernandez

Tillåtna hjälpmedel/Allowed aids: Räknare/Calculator

Maximum score: 38

<u>Grade</u>	<u>Score</u>
Fail:	Below 16
Grade 3:	At least 16
Grade 4:	At least 23
Grade 5:	At least 31

Write your answers in a structured and understandable way. Motivate your answers!

Good luck!

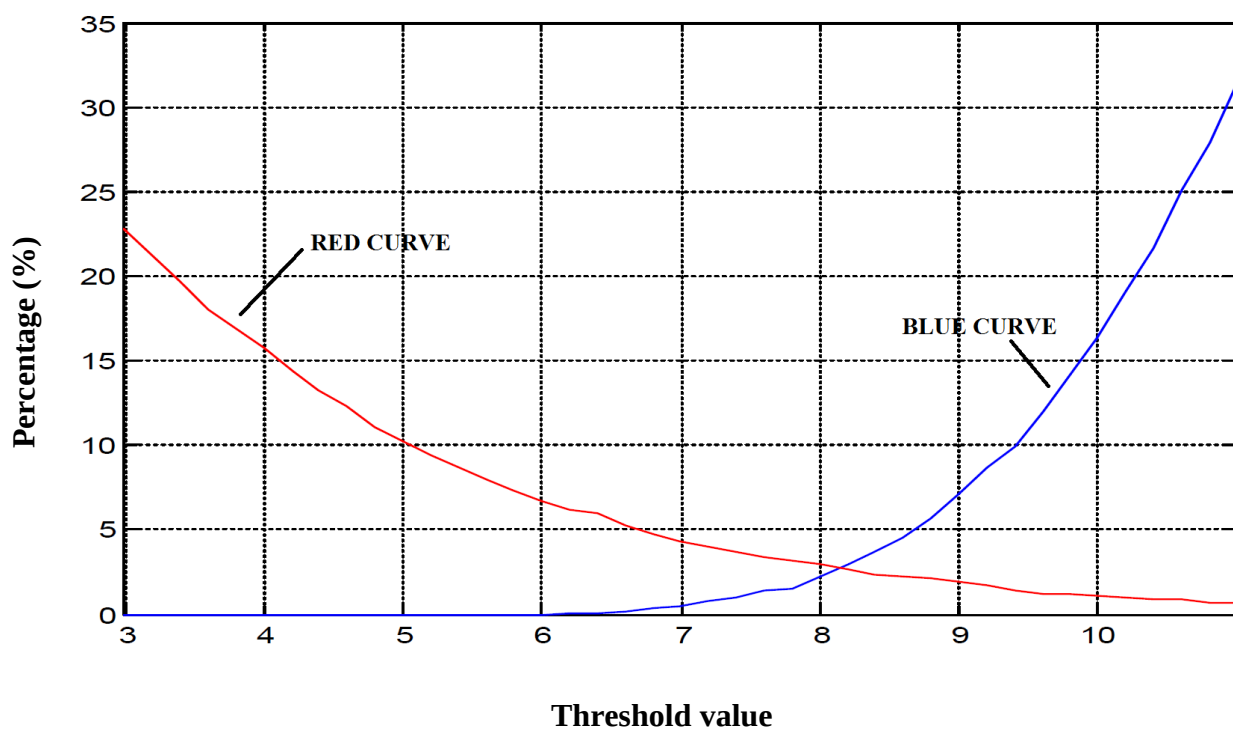
Exercise 1. (6p)

- Indicate at least three requirements/properties that a human characteristic should satisfy in order to be useful to be employed as a biometric trait (2p)
- Consider the following biometrics: fingerprint, face and voice. For each of them, indicate if:
(H=high) possess the mentioned characteristic to a high extent
(L=low) possess the mentioned characteristic to a few extent
(M=medium) something in the middle
using the three properties that you indicated in a). (2p)
- Choose two properties of a) and indicate why they are important and how they affect the performance of a biometric system (2p)

Exercise 2. (6p)

A biometric system employs *distance* as comparison measure for matching biometric samples. A *small* value of the distance means *high similarity between the samples*, while a *high* value of the distance means *that the samples are very different*.

The figure below shows how the two types of errors (FAR and FRR) vary for different thresholds when the system operates in verification mode.



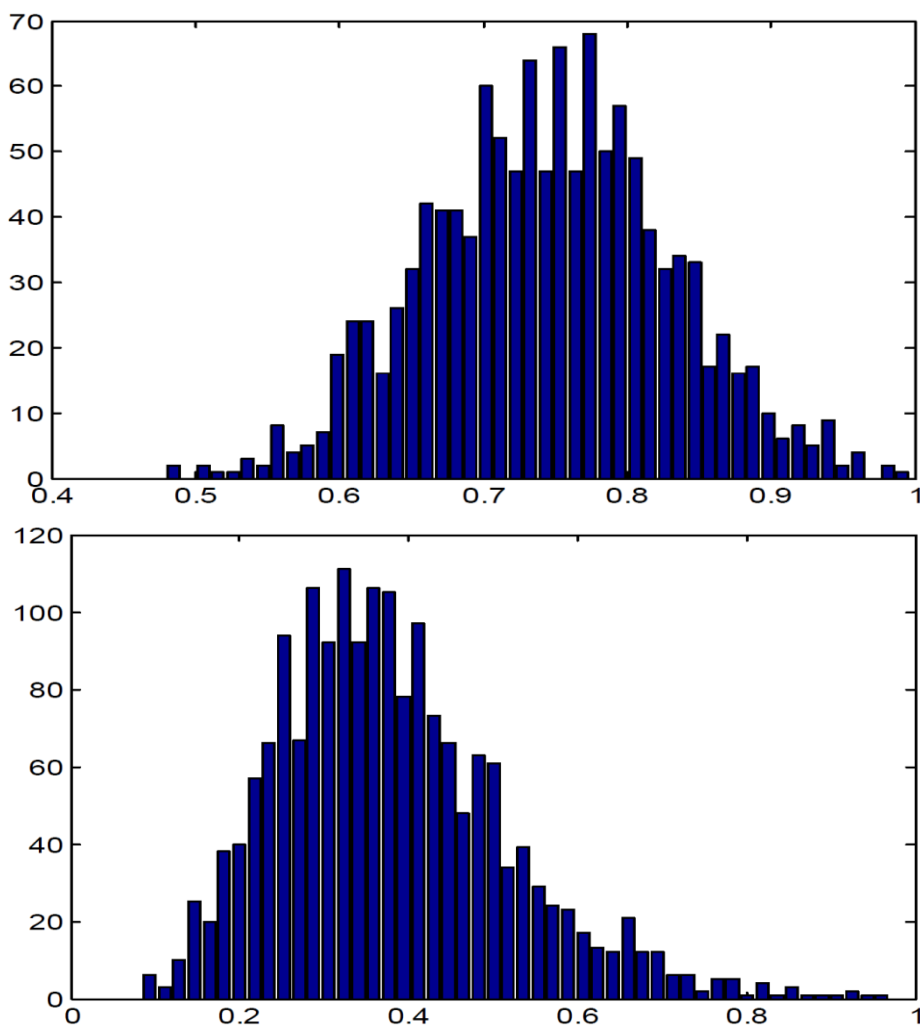
- Indicate which curve (red and blue) corresponds to the FAR and FRR. Motivate your answer (2p)
- We want the system to operate as a high convenient system. Choose an appropriate threshold and indicate the FAR and FRR values at that threshold (both FAR and FRR should not exceed 10%). Motivate your choice. (2p)
- From the above graph, create the ROC curve of the system. Use the x-axis for FAR and the y-axis for FRR. Use at least five different thresholds. (2p)

Exercise 3. (12p)

A biometric system is working in *verification* mode. We carry out different *genuine* and *impostor* tests. In concrete, 1200 matchings of samples from the same person (genuine) and 1800 matchings of samples from different persons (impostor) are done. In each matching, we employ a *similarity* measure, so *the higher the value of the measure, the more similar are the samples*.

Results of the experiments are reported in the following histograms. The first histogram is when samples from the same person are matched (genuine), and the second histogram is when samples from different persons are matched (impostor). The x-axis of the histogram indicates the similarity value, and the y-axis indicates the number of matchings. A table with results from the histograms is also reported.

- a) Calculate the FAR and FRR of the system for different thresholds of the similarity measure. Report you results both in
- a table, indicating the threshold and the respective errors (employ at least 10 different thresholds, with at least 6 different thresholds where the two histograms overlap)
 - a graph (draw errors in the range 0-20% approximately), indicating clearly in the axes the values of each point
- (9p)
- b) Indicate the threshold of the EER (Equal Error Rate) and the value of the EER. Mark the values also in your graph of a). (1p)
- c) The system has the requirement of maximum 1% FAR when FRR=10%. Based on your results of a), can the system meet this requirement? Motivate your answer. (2p)



Genuine			Impostor	
<i>Quantity</i>	<i>Similarity</i>		<i>Quantity</i>	<i>Similarity</i>
2	0.4851		6	0.0932
0	0.4954		3	0.1109
2	0.5057		10	0.1286
1	0.5161		25	0.1463
1	0.5264		20	0.1640
3	0.5367		38	0.1817
2	0.5471		40	0.1994
8	0.5574		57	0.2170
4	0.5677		66	0.2347
5	0.5780		94	0.2524
7	0.5884		67	0.2701
19	0.5987		106	0.2878
24	0.6090		92	0.3055
24	0.6194		111	0.3232
16	0.6297		92	0.3409
26	0.6400		106	0.3586
32	0.6503		105	0.3763
42	0.6607		78	0.3940
41	0.6710		97	0.4117
41	0.6813		73	0.4294
37	0.6917		66	0.4471
60	0.7020		48	0.4648
52	0.7123		63	0.4825
47	0.7227		61	0.5002
64	0.7330		34	0.5179
47	0.7433		39	0.5356
66	0.7536		29	0.5533
47	0.7640		24	0.5710
68	0.7743		23	0.5887
50	0.7846		17	0.6064
57	0.7950		13	0.6241
49	0.8053		12	0.6418
38	0.8156		21	0.6595
32	0.8259		12	0.6772
34	0.8363		12	0.6949
33	0.8466		6	0.7126
17	0.8569		6	0.7303
22	0.8673		2	0.7480
16	0.8776		5	0.7657
17	0.8879		5	0.7834
10	0.8983		1	0.8011
6	0.9086		4	0.8188
8	0.9189		1	0.8365
5	0.9292		3	0.8541
9	0.9396		1	0.8718
2	0.9499		1	0.8895
4	0.9602		1	0.9072
0	0.9706		2	0.9249
2	0.9809		1	0.9426
1	0.9912		1	0.9603

Exercise 4. (6p)

A biometric system works in *identification* mode. We carry out a number of tests, with the results shown below. For each test, we obtain a list of ID numbers, sorted by the most similar. The test has been carried out for a total of 15 people, with 3 tests for each person. The individuals have ID numbers: 1, 2, 3, ..., 15.

Each line represents a sorted list of ID numbers. Lines are shown in groups of 3, representing the 3 tests of each person: the first group corresponds to ID=1, the next group corresponds to ID=2, and so on.

a) Calculate:

- the function $P(k)$ that indicates the probability that the correct identity is given in position k of the sorted list.
- the function $\text{TPIR}(M)$ that indicates the probability that the correct identity is given in the first M positions of the sorted list.

Show your calculations of $P(k)$ and $\text{TPIR}(M)$ both in a table and a graph. (4p)

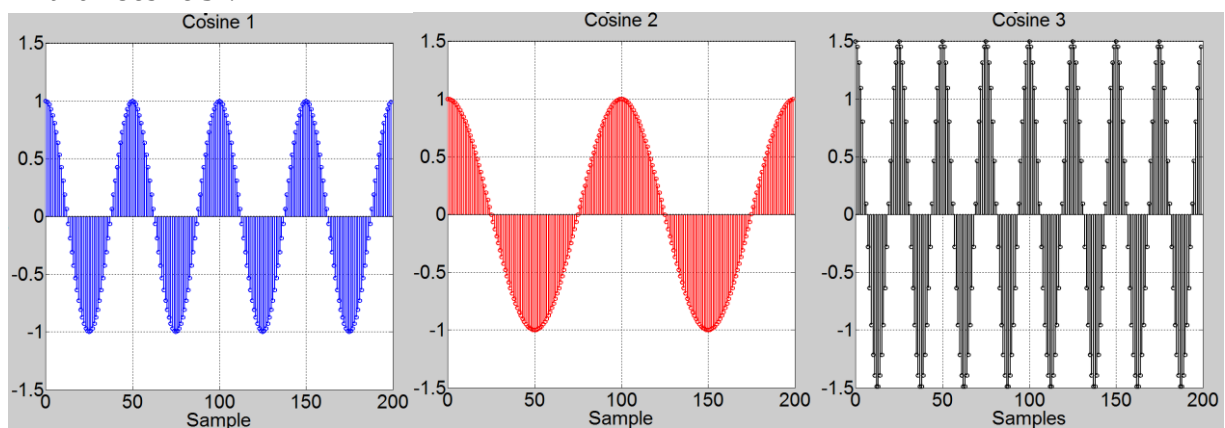
b) The system has the requirement that the probability that the correct identity is given within the first 3 positions is at least 80%. Does the system meet this requirement? Motivate your answer! (2p)

1	5	2	10	4	13	12	11	3	6	9	7	15	14	8
4	1	3	10	12	9	7	6	8	14	11	5	15	13	2
1	5	9	11	8	13	14	2	3	12	15	7	4	10	6
7	2	11	5	4	9	3	1	14	15	10	6	12	13	8
2	12	8	3	6	14	9	13	4	11	5	10	15	7	1
8	10	6	2	3	14	7	12	13	15	1	11	9	4	5
3	15	11	12	14	4	2	6	7	1	10	8	9	5	13
15	3	7	4	14	10	1	11	5	8	12	9	6	2	13
3	1	10	6	5	11	15	9	8	13	7	14	4	12	2
3	4	1	8	10	14	15	5	6	2	11	13	12	7	9
2	8	11	6	15	4	5	1	9	13	12	3	14	7	10
4	6	8	1	11	12	3	5	10	15	2	14	9	7	13
5	12	1	15	14	7	9	3	6	2	13	4	10	11	8
15	14	5	1	12	9	6	11	7	10	13	8	4	3	2
5	9	2	7	4	3	10	13	15	11	1	12	6	14	8
8	4	6	9	11	3	7	12	10	2	13	5	14	15	1
3	1	10	6	11	12	8	2	14	5	15	4	13	7	9
6	3	5	9	12	14	2	1	11	7	10	15	8	13	4
7	3	13	9	15	11	6	1	2	5	10	14	4	8	12
7	13	5	1	14	10	15	3	4	2	8	11	6	12	9
7	10	4	2	9	1	11	6	5	13	12	14	3	15	8
9	8	12	6	7	11	15	1	14	4	3	2	13	5	10
8	14	10	9	5	13	15	3	7	2	12	6	1	4	11
10	15	8	13	11	14	6	3	7	2	1	4	12	5	9

9	12	15	13	2	4	3	6	1	8	7	10	5	11	14
13	5	12	9	11	3	15	2	7	1	8	4	10	14	6
4	9	11	5	12	1	15	10	2	7	3	14	8	13	6
11	10	14	9	5	15	12	1	6	13	2	8	3	7	4
6	11	10	7	2	3	1	5	8	9	15	14	13	12	4
10	6	1	11	5	8	12	3	15	13	14	4	9	2	7
14	9	6	13	11	5	10	15	12	4	2	8	1	7	3
10	5	3	4	11	2	12	1	13	15	14	8	9	7	6
11	3	12	2	13	15	7	4	5	10	8	6	1	14	9
10	5	2	13	4	9	14	6	15	12	1	8	7	11	3
2	5	12	1	9	6	8	7	14	15	3	10	13	11	4
12	14	9	1	15	10	5	6	11	8	2	7	13	4	3
13	14	5	10	15	7	8	9	1	12	4	6	2	3	11
13	6	10	3	2	9	5	15	7	4	12	1	8	14	11
5	13	15	11	3	1	9	14	12	7	4	2	8	6	10
14	3	5	10	12	2	13	6	11	7	1	9	8	15	4
5	3	14	6	2	4	1	10	8	7	13	9	11	12	15
14	3	11	2	6	12	5	1	4	13	7	9	10	8	15
4	3	5	14	11	15	13	7	9	8	2	12	10	6	1
15	9	2	11	5	6	13	8	10	1	7	4	14	12	3
15	3	7	13	6	2	5	11	4	8	1	9	10	12	14

Exercise 5.(8p)

- a) The following figures show 200 samples of three cosine signals, named "Cosine 1", "Cosine 2", and "Cosine 3".



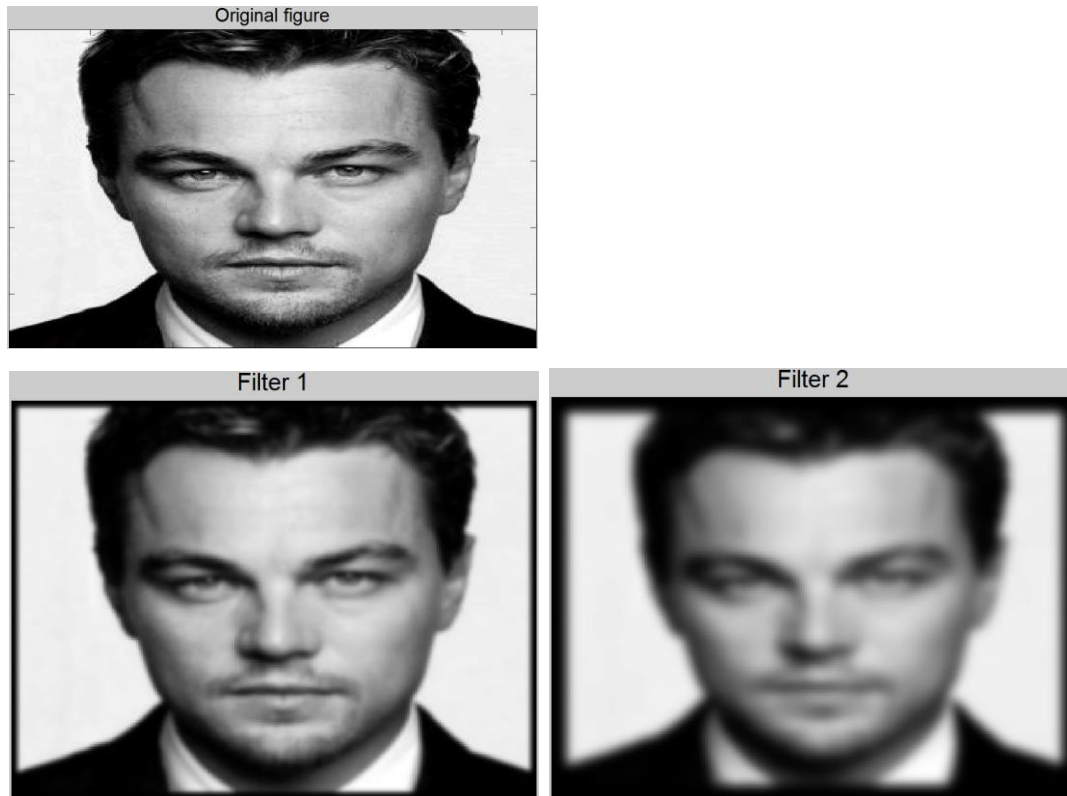
Indicate:

- The signal(s) with the highest frequency and the signal(s) with the smallest frequency
- The signal(s) with the highest amplitude and the signal(s) with the smallest amplitude

Motivate your answers! (3p)

- b) Two Gaussian filters are applied to a picture, with the results shown below. Each filter has a different size (given by the parameter “s” or “σ” of the Gaussian function):

$$G(x, y) = \exp\left(-\frac{(x^2 + y^2)}{2\sigma^2}\right)$$



Describe the effect on the image of a Gaussian filter.

Indicate which of the two filters (Filter1 or Filter2) has higher size.

Motivate your answers! (3p)

- c) The following figures show the extraction of S and M points of a fingerprint before and after applying enhancement with a Gaussian filter.

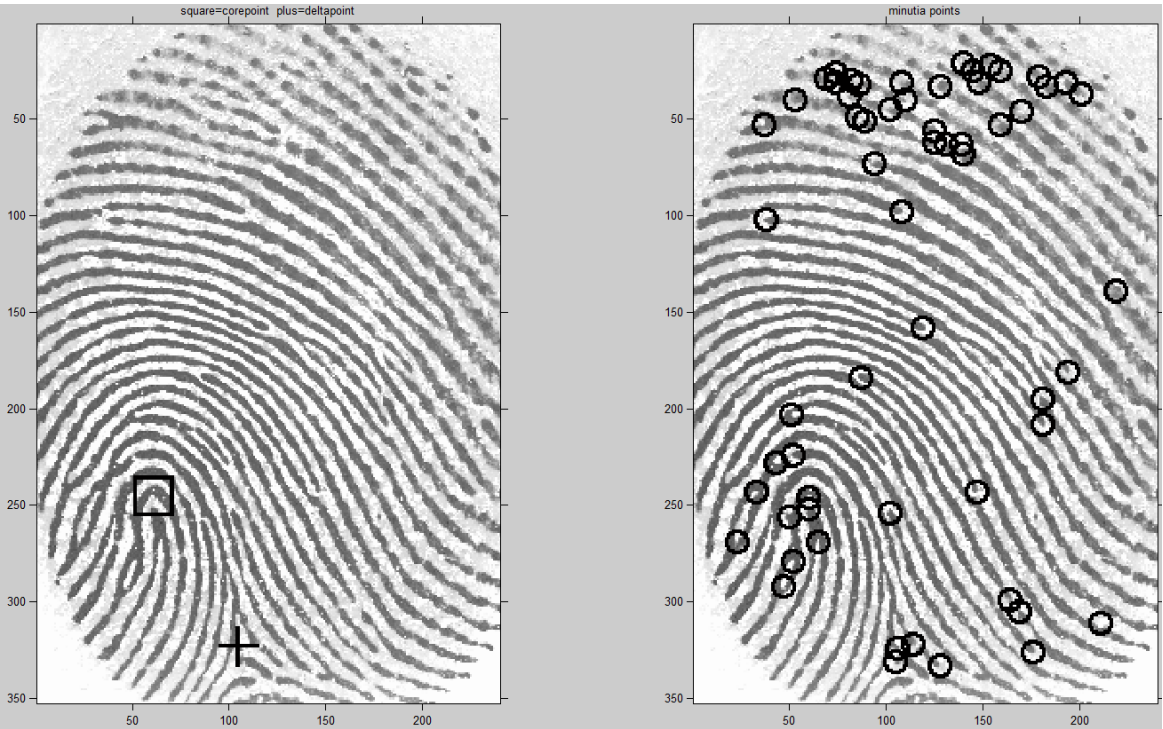
Indicate the parts of the fingerprint that have poor quality.

Indicate if, in your opinion, the enhancement is successful.

Justify your answer both by referring to the lines of the fingerprint before/after enhancement, and to the M points detected without and with enhancement.

Motivate your answers! (2p)

Before enhancement:



After enhancement:

